# XGBOOST-BASED URL PHISHING DETECTION METHOD WITH CROSS-DATASET VALIDATION

**SEMINAR REPORT**

Submitted by

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**St. Joseph's College of Engineering and Technology, Palai**

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# CERTIFICATE

This is to certify that the seminar report entitled **"XGBOOST-BASED URL PHISHING DETECTION METHOD WITH CROSS-DATASET VALIDATION"** submitted by **DEON GEORGE (SJC23CS086)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide report of the seminar presented by him under my guidance and supervision.

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# ABSTRACT

Phishing URLs are frequently used to deceive users into revealing sensitive information, transferring money, or accessing malicious resources. Machine-learning-based URL detection can provide early warnings, but models evaluated on a single dataset may not generalise well to URLs collected from different sources. This seminar reviews X-PHIDE, an XGBoost-based URL phishing detection method proposed by Misiek and Hyla. The study evaluates a custom dataset containing 75,738 URLs, the 639,723-URL GramBeddings dataset, and a feature-compatible subset of PhiUSIIL containing 47,103 URLs. More than 90 candidate URL and domain features are analysed, and eight features are selected using cross-dataset importance and stability criteria. Bayesian hyperparameter optimisation is used to tune XGBoost, and the system combines allowlist/blocklist checks with model predictions in a browser plug-in workflow. The reported model achieves 90.72% accuracy and 91.16% F1 on the custom test set, while F1 scores on GramBeddings and PhiUSIIL are 73.25% and 69.37%, respectively. In a comparison with a neural network, XGBoost trains 12.6 times faster and uses 3.6 times less memory, although the neural network has faster inference and a smaller model size. The results demonstrate the value of efficient tree-based detection and, importantly, show why cross-dataset validation is necessary. Continued monitoring, model updates, and independent testing remain important for practical deployment. [1]

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# Chapter 1

# Introduction

## 1.1 Overview of URL Phishing

Phishing is a social-engineering attack in which an attacker deceives a user into revealing sensitive information, transferring money, or approving an unintended action. A common first step is to distribute a deceptive Uniform Resource Locator (URL) that directs a victim to a fraudulent website or to a resource associated with malware. Such links may be shared through email, messaging applications, social networks, or compromised websites. Successful attacks can lead to financial loss, identity theft, and reputational damage. Malicious URLs may be associated with spam, phishing, malware distribution, or website defacement. Although these threats are not identical, many detection systems reduce the decision to a binary classification: benign or malicious. [1]

## 1.2 Motivation and Importance

Browser-based protection needs to assess links quickly, including URLs that have not previously appeared in public blocklists. Static lists are valuable for known threats but cannot by themselves recognise every newly created or modified malicious URL. Machine-learning models can identify patterns in URL characteristics, but their performance may decline when the source or distribution of URLs changes. This is known as dataset shift or a generalisation gap.

The selected paper, by Misiek and Hyla, introduces X-PHIDE, an XGBoost-based phishing URL detection method and evaluates it across three datasets. Its central emphasis is not only

high performance on a single test split, but also cross-dataset validation, feature stability, and practical computational cost. [1]

## 1.3 Objectives

The objectives of this seminar report are:

- To explain the challenges involved in detecting phishing URLs and in obtaining reliable generalisation across different datasets.
- To summarize the dataset preparation, URL feature extraction, feature-selection process, and XGBoost model described in the base paper.
- To explain the cross-dataset validation strategy, Bayesian hyperparameter optimisation, and evaluation measures used by the authors.
- To review the paper's reported detection performance, computational requirements, and comparison with neural-network and baseline models.
- To discuss the potential browser-based applications, limitations, and future research directions of the X-PHIDE method.

## 1.4 Scope of the Report

This report is a technical review of the selected research paper and its reported experiments. It explains the datasets, URL features, model pipeline, validation procedure, results, and browser plug-in workflow as presented by the authors. The report does not claim to reproduce the training experiments or to implement a complete phishing-detection system independently. All numerical results are attributed to the original paper. [1]

# Chapter 2

# Background

## 2.1 URL-Based Phishing Detection

A URL contains structural and lexical information that can be used to estimate whether it is suspicious. Examples include total URL length, the number of subdomain components, the presence of brand-related keywords, authentication-related keywords, nameserver counts, DNS information, HTTPS presence, and domain registration age. These signals are imperfect individually: legitimate and malicious URLs can share similar characteristics, and their usefulness may vary across datasets. [1]

## 2.2 Machine Learning for Structured Features

The paper uses Extreme Gradient Boosting (XGBoost), an ensemble method that builds decision trees sequentially to improve predictions. XGBoost is suited to structured/tabular features, supports missing values, and provides feature-importance information. These properties make it attractive for security applications where training efficiency and an interpretable feature-level view are useful. The paper compares its performance with a neural network, noting that the neural network can have faster inference and a smaller model despite requiring more training time in the reported experiment. [1]

## 2.3 Cross-Dataset Validation

A random train/test split from one dataset can overestimate real-world performance if both partitions share similar collection practices, URL structures, or temporal patterns. Cross-dataset validation evaluates a model on data drawn from a different source. It exposes whether learned patterns transfer to different URL populations. Prior work discussed in the paper reports F1-score decreases of 10–32% under dataset shift, motivating evaluation beyond a single dataset. [1]

## 2.4 Feature Importance and Stability

Feature importance estimates which input variables contribute to a model's decisions. However, a feature can be highly predictive in one dataset and weak in another because collection methods and class distributions differ. X-PHIDE therefore examines both predictive importance and stability across datasets, seeking features that remain useful in different environments. Feature importance is an explanatory signal, not proof of causation.

# Chapter 3

# Problem Statement

## 3.1 Problem Addressed

Phishing URL detectors may achieve strong results when training and testing use the same dataset, yet perform less reliably on URLs from other sources. This limits confidence in deploying such models as browser protection. URL features can also vary in availability and predictive value across datasets, making feature selection a key part of generalisation.

**Problem statement:** How can an XGBoost-based URL phishing detector be designed and evaluated to balance detection quality, computational efficiency, and generalisation across distinct URL datasets?

## 3.2 Research Questions

The paper investigates three questions: (1) whether XGBoost achieves cross-dataset generalisation compared with baseline methods; (2) how feature-importance stability relates to cross-dataset performance; and (3) how the computational efficiency of XGBoost compares with neural networks for real-time deployment. [1]

## 3.3 Scope of the Seminar

The scope of this seminar includes the following aspects of the published X-PHIDE method:

- The motivation for URL-only phishing detection and the challenges caused by dataset shift and changing URL distributions.
- The sources, composition, and role of the custom, GramBeddings, and PhiUSIIL datasets used in the study.
- The extraction of lexical, structural, domain-related, and other URL-based features, together with the criteria used to select stable features across datasets.
- The XGBoost classification pipeline, Bayesian hyperparameter optimisation, decision thresholds, and comparison with baseline and neural-network models.
- The evaluation measures used to assess classification quality, cross-dataset generalisation, training and inference time, and memory requirements.
- The proposed browser plug-in workflow, including allowlist and blocklist checks and the presentation of URL risk categories to users.
- The reported strengths, limitations, practical deployment considerations, and possible directions for future research.

The report focuses on explaining and analysing the methods and results presented by Misiek and Hyla. It does not include independent data collection, model training, statistical re-analysis, or implementation of a complete production-ready browser extension. All numerical results and performance claims are attributed to the original authors; no independent reproduction is claimed. [1]

# Chapter 4

# Literature Review

## 4.1 GramBeddings

Bozkir, Dalgiç, and Aydos propose GramBeddings, a neural-network method for identifying phishing web pages from URLs. The method represents URL text using character-level n-grams and learned embeddings, allowing the model to capture local patterns in strings such as unusual tokens, word fragments, and character sequences. The work demonstrates the value of neural representations for URL-only phishing detection on a large dataset. [2]

The paper is relevant to Misiek and Hyla's base paper because GramBeddings is one of the datasets used in the X-PHIDE cross-dataset evaluation. It also provides a meaningful neural comparison for X-PHIDE: both methods use only URL-derived information, but X-PHIDE emphasises a small set of stable, engineered features and the efficiency of XGBoost, whereas GramBeddings learns representations directly from URL text. [1, 2]

## 4.2 Unsupervised Domain Adaptation

Rashid, Doyle, Han, and Seneviratne study the generalisation of phishing URL detection using unsupervised domain adaptation. Their work addresses the situation in which labelled training data are available in a source domain but the target domain has different characteristics and limited or no labels. The approach aims to reduce the effect of domain shift so that a detector can transfer more reliably between URL datasets. [3]

This paper is directly relevant to the base paper because both focus on performance across

changing data distributions rather than performance on a single random split. X-PHIDE does not primarily adapt a model with unlabelled target data; instead, it investigates cross-dataset performance, selects features that remain useful across datasets, and evaluates the resulting XGBoost model on independent sources. Domain adaptation therefore provides a complementary strategy and reinforces the base paper's motivation for testing generalisation explicitly. [1, 3]

## 4.3 Survey of Phishing Environments and Countermeasures

Aleroud and Zhou present a survey of phishing environments, attack techniques, and countermeasures. The paper organises the broader phishing problem by examining how attacks are delivered, how users are deceived, and how technical and organisational controls can reduce the threat. It provides background for understanding why detection systems must consider changing attacker behaviour and diverse sources of evidence. [4]

The survey is relevant to the base paper as a broad security context for URL-based detection. While it covers phishing countermeasures more widely than X-PHIDE, its discussion of varied attack environments supports the base paper's concern that a detector trained under one collection condition may not generalise to another. X-PHIDE addresses this concern more narrowly through URL-only features, cross-dataset validation, and a browser plug-in workflow. [1, 4]

## 4.4 Machine Learning and Deep Learning Review

Divakaran and Oest review the use of machine learning and deep learning for phishing detection. The review compares families of detection methods, input features, learning architectures, and evaluation practices, showing how conventional machine-learning models and neural networks have both been applied to phishing-related classification tasks. It also highlights the importance of selecting appropriate data and evaluation settings when comparing reported results. [5]

This review is relevant to the base paper because it places X-PHIDE among the larger set of machine-learning and deep-learning approaches to phishing detection. The base paper follows the review's broad comparison perspective but focuses on a specific unresolved issue: whether strong results transfer across independent URL datasets. Its comparison of XGBoost with a neural model, together with measurements of training time, inference time, and memory use,

gives the general review criteria a concrete application. [1, 5]

## 4.5 Hybrid DNN–LSTM Detection

Özcan, Catal, Dönmez, and Şentürk propose a hybrid DNN–LSTM model for detecting phishing URLs. The model combines dense neural-network processing with LSTM-based sequence modelling so that both feature relationships and sequential patterns in URL data can be learned. This represents a more complex neural alternative to traditional classifiers and illustrates how URL detection can use learned representations rather than relying only on manually selected features. [6]

The hybrid model is relevant to the base paper because Misiek and Hyla compare X-PHIDE with a neural-network approach while considering practical deployment costs. The two studies therefore address a similar detection task with different modelling priorities: the DNN–LSTM work emphasises neural sequence modelling, while X-PHIDE emphasises stable cross-dataset features, XGBoost classification, and browser-level efficiency. This comparison helps justify evaluating accuracy together with computational requirements. [1, 6]

# Chapter 5

# Technology and Paper Overview

## 5.1 Selected Paper

M. Misiek and T. Hyla, "XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation," *IEEE Access*, vol. 14, 2026, pp. 43200–43212, doi: 10.1109/ACCESS.2026.3672690. The article was published online on 10 March 2026. [1]

## 5.2 Objectives

- Develop an XGBoost-based method to classify potentially malicious URLs.
- Evaluate the model on datasets from different sources rather than relying on a single split.
- Identify URL features that provide useful and stable signals across datasets.
- Tune the model using Bayesian hyperparameter optimisation.
- Compare detection metrics and computational characteristics with a neural network and baseline classifiers.
- Describe a browser plug-in workflow that labels URLs as safe, suspicious, or potentially harmful.

## 5.3 Datasets

The authors use three datasets. The custom dataset contains 75,738 URLs assembled from PhishTank, AdaURL, the Kaggle Phishing URLs Dataset by Hassaan Mustafavi, and PhishDataset. GramBeddings contains 639,723 URLs and is used for feature analysis and cross-dataset evaluation. PhiUSIIL provides an independent evaluation set; the feature-compatible subset used in the experiments contains 47,103 URLs. These datasets differ in scale and URL characteristics, which makes them useful for studying generalisation. [1]

## 5.4 Proposed Pipeline

The high-level pipeline is: URL input; allowlist/blocklist lookup; extraction of more than 90 candidate features; cross-dataset analysis and selection of eight stable features; Bayesian hyperparameter optimisation; XGBoost prediction; risk classification; and display of the result in a browser extension. Users may also confirm classifications, allowing domains to be added to the relevant allowlist or blocklist. [1]

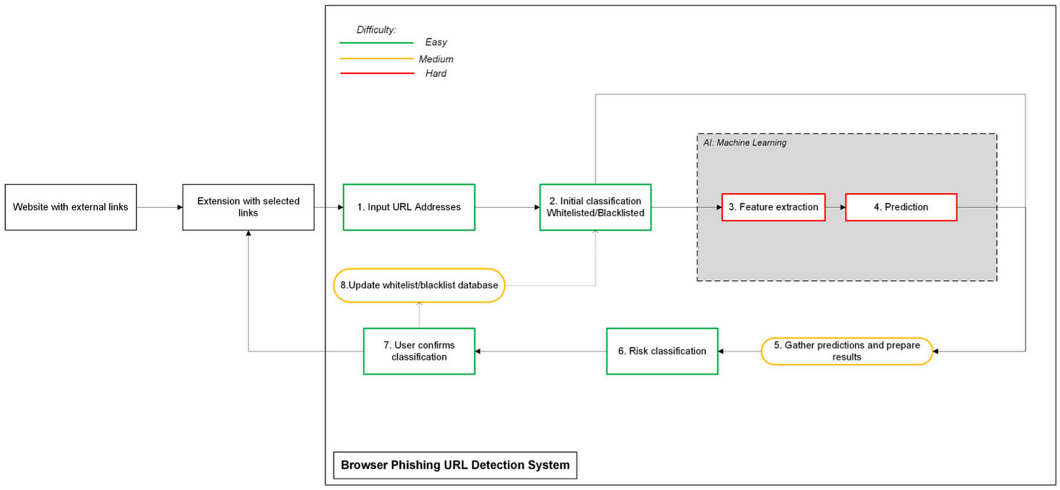



**FIGURE 1.** **System overview of the X-PHIDE phishing detection method.**

Figure 5.1: System overview of the browser-based X-PHIDE phishing detection method.
[fig:system-overview]

## 5.5 Main Contributions

The main contribution is the evaluation of URL phishing detection under dataset shift. The paper also examines feature-importance stability, reports dataset-specific results and thresholds, and compares XGBoost with a neural network in terms of accuracy, training time, inference time, and memory use.

# Chapter 6

# Working Principle

## 6.1 URL Input and List Check

When a user clicks an external link, the browser plug-in captures the destination URL. It first checks whether the domain appears in a trusted allowlist or a known-malicious blocklist. If a domain is already listed, the corresponding classification can be returned without running the machine-learning model. If the domain is not listed, it proceeds to feature extraction and prediction. [1]

## 6.2 Feature Extraction

The research system extracts more than 90 candidate URL and domain characteristics. These include lexical features, DNS and network information, security/authentication signals, domain reputation, and URL behaviour. Examples include URL length, the number of hyphens in the URL path, brand-related keywords, authentication-related keywords in subdomains, nameserver count, the number of dots in the URL, HTTPS presence, and domain registration age. Since feature availability differs between datasets, common features are compared before selecting the final set. [1]

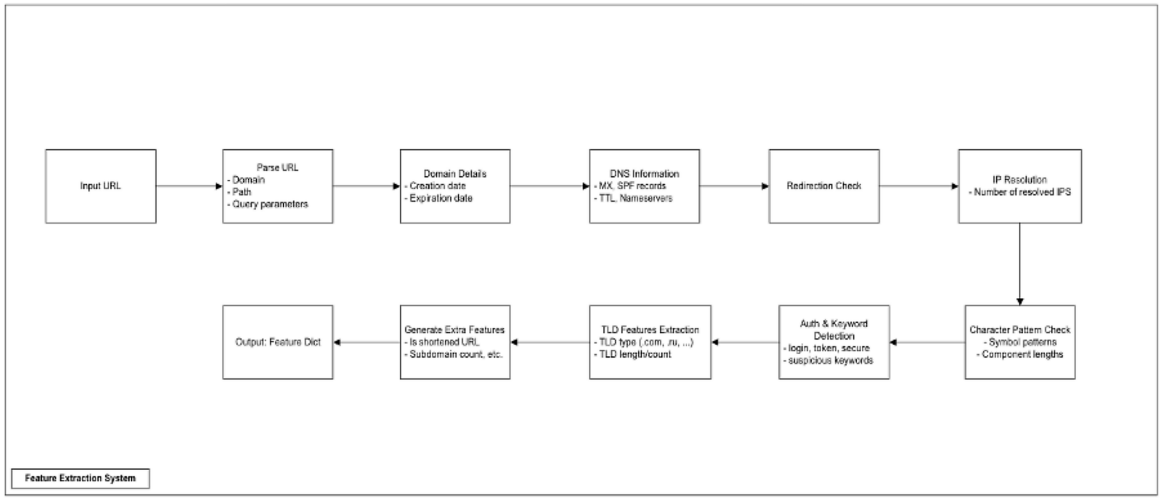



**FIGURE 4.** **Feature extraction method.**

Figure 6.1: Feature extraction pipeline used to construct URL and domain features. [fig:feature-pipeline]

## 6.3 Selection of Eight Features

The authors apply criteria intended to prioritise cross-dataset reliability over performance on one dataset alone: mean importance greater than 0.02, minimum importance greater than 0.005, coefficient of variation below 0.8, and presence in at least 75% of datasets. The paper notes that only one of 32 analysed features met every criterion, illustrating the difficulty of finding stable signals. The final model uses eight selected features, including URL length, hyphen count in the path, brand-related keywords, authentication-related subdomain count, nameserver count, dot count, HTTPS presence, and domain registration age. [1]

## 6.4 Bayesian Hyperparameter Optimisation

The model's hyperparameters were tuned using BayesSearchCV from scikit-optimize. The search used stratified five-fold cross-validation on the custom dataset, a budget of 100 evaluations, and parallel execution across available CPU cores. The optimisation objective was cross-validated accuracy. The authors then refitted the selected configuration and evaluated it on held-out and independent datasets. Class imbalance was addressed using XGBoost's `scale_pos_weight`, tuned to approximately 1.95. [1]

## 6.5 Prediction and Risk Classification

The trained XGBoost model estimates the probability that a URL is associated with phishing. In the described plug-in workflow, an allowlisted domain is marked safe; a blocklisted domain is marked potentially harmful. For domains not on either list, probabilities below 40% are labelled safe, values from 40% to 80% suspicious, and values above 80% potentially harmful. The article also reports experimental evaluation at a fixed threshold of 0.85 and threshold selection to meet target specificity levels; these experiment thresholds should not be confused with the user-facing three-category rule. [1]

## 6.6 Result Display and Feedback

The plug-in displays the prediction and risk category to the user. Users can manually confirm a classification, and the domain decision is saved to the corresponding allowlist or blocklist for future checks. This combination of lists and model predictions provides a practical route to real-time URL screening.

## 6.7 Step-by-Step Summary

1. Capture the URL when an external link is selected.
2. Check the domain against allowlist and blocklist data.
3. Extract the required URL and domain features if no list entry exists.
4. Apply the selected eight-feature XGBoost model.
5. Convert the prediction into a risk category.
6. Show the classification and allow user feedback.
7. Store confirmed domain classifications for future reference.

# Chapter 7

# Key Findings from the Paper

**Attribution:** All numerical results in this chapter are reported by Misiek and Hyla in the selected paper; they were not independently reproduced for this seminar. [1]

## 7.1 Custom Dataset Performance

On the held-out custom test dataset, the optimised XGBoost model achieved 90.72% accuracy, 94.6% recall, 86.8% specificity, 88.0% precision, and an F1 score of 91.16%. The recall indicates that most phishing URLs in this test set were detected, while specificity reflects the proportion of legitimate URLs correctly identified as non-phishing. [1]

## 7.2 Cross-Dataset Generalisation

| **Evaluation dataset** | **F1 score** |
|---|---|
| Custom test dataset | 91.16% |
| GramBeddings | 73.25% |
| PhiUSIIL | 69.37% |
| Average across three datasets | 77.93% |

Table 7.1: Reported cross-dataset F1 scores for XGBoost.
[tab:cross_dataset_f1]

The results show a substantial performance decline on independent data compared with the custom test set. The average F1 score across the three datasets is 77.93%. This supports

the paper's central point: high performance on one dataset does not guarantee equally strong performance on another. [1]

**TABLE 14. Cross-dataset performance of XGBoost with 95% bootstrap confidence intervals over 10 repeats (fixed threshold 0.50). Values are reported as mean [$CI_{2.5\%}$, $CI_{97.5\%}$].**

| Dataset | Accuracy | F1 | ROC-AUC |
|---|---|---|---|
| Custom Test | 0.9084 [0.9076, 0.9093] | 0.9131 [0.9123, 0.9139] | 0.9759 [0.9755, 0.9763] |
| GramBeddings Test | 0.6468 [0.6440, 0.6498] | 0.7292 [0.7266, 0.7318] | 0.7580 [0.7549, 0.7610] |
| PhiUSIIL Test | 0.6221 [0.6172, 0.6270] | 0.6695 [0.6675, 0.6716] | 0.8050 [0.8026, 0.8072] |

CI details: percentile bootstrap CI over repeats (resamples=2000). Each repeat refits the model and evaluates on the same held-out test split for the given dataset.

Table 7.2: Cross-dataset performance of XGBoost with 95% bootstrap confidence intervals over 10 repeats.
[tab:dataset-comparison]

## 7.3 Comparison with a Neural Network

In the reported benchmark, XGBoost achieved 2.1% higher accuracy, trained 12.6 times faster (0.95 seconds compared with 12.01 seconds), and used 3.6 times less memory. The neural network had faster inference (7.51 ms compared with 20.6 ms) and a smaller model size. The best choice therefore depends on whether training cost, inference latency, memory footprint, or predictive quality is the primary constraint. [1]

**TABLE 15. Paired sign-flip permutation tests across repeats for $F_1$ (two-sided; permutations = 20000; reference = XGBoost). $\Delta F_1$ is computed as (model − XGBoost).**

| Dataset | Model | $\Delta F_1$ | p-value |
|---|---|---|---|
| Custom Test | LightGBM | −0.000983 | 0.0136 |
| | RandomForest | −0.007408 | 0.0018 |
| | CatBoost | −0.012365 | 0.0023 |
| | MLP | −0.015809 | 0.0017 |
| | LogisticRegression | −0.193753 | 0.0018 |
| GramBeddings Test | MLP | +0.018289 | 0.0018 |
| | RandomForest | +0.011746 | 0.0023 |
| | LightGBM | +0.002546 | 0.1183 |
| | CatBoost | −0.001806 | 0.2015 |
| | LogisticRegression | −0.048617 | 0.0019 |
| PhiUSIIL Test | RandomForest | +0.007359 | 0.0040 |
| | LightGBM | −0.003627 | 0.0062 |
| | CatBoost | −0.011745 | 0.0019 |
| | MLP | −0.002931 | 0.6398 |
| | LogisticRegression | −0.057107 | 0.0020 |

Note: Results are based on $n = 10$ paired repeats per dataset. $p$-values are uncorrected for multiple comparisons and are reported to support the robustness of cross-dataset differences requested by reviewers.

Table 7.3: Reported paired sign-flip permutation tests comparing models across datasets.
[tab:model-comparison]

## 7.4 Impact of HTTPS

The contribution of HTTPS presence differed by dataset. The reported accuracy change when the feature was included was +4.97 percentage points for the custom dataset, +0.04 points for GramBeddings, and +12.81 points for PhiUSIIL. The paper explains that the value of HTTPS as a signal depends on the distribution of legitimate and phishing URLs in each dataset. HTTPS alone should therefore not be treated as proof that a website is safe. [1]

## 7.5 Summary

XGBoost performed strongly on the custom test set and was computationally efficient to train, but the independent-dataset results exposed a meaningful generalisation gap. Stable feature selection and cross-dataset evaluation are essential for estimating how a detector may behave outside its training environment.

# Chapter 8

# Results and Analysis

## 8.1 Feature Stability and Dataset Differences

The article compares feature importance across the custom, GramBeddings, and PhiUSIIL datasets. URL length was identified as a relatively robust feature, with reported mean importance of 0.0319 and coefficient of variation of 0.7843. HTTPS showed much greater variation in importance, from 0.125 in GramBeddings to 0.513 in PhiUSIIL. This is consistent with the observed dataset-dependent effect of HTTPS on accuracy. [1]

## 8.2 Why Dataset Shift Matters

Datasets can differ in URL length distributions, HTTPS adoption, phishing prevalence, collection period, and source. A classifier may inadvertently learn patterns that distinguish the labels in one dataset but do not persist in another. Cross-dataset tests reveal this risk and give a more realistic view of the model's limits than a single random split.

## 8.3 Evaluation Metrics

Accuracy measures the proportion of all samples classified correctly. Recall measures the proportion of phishing URLs detected. Specificity measures the proportion of legitimate URLs correctly identified. Precision measures how many URLs flagged as phishing are actually phishing, and the F1 score balances precision and recall. No single metric fully describes

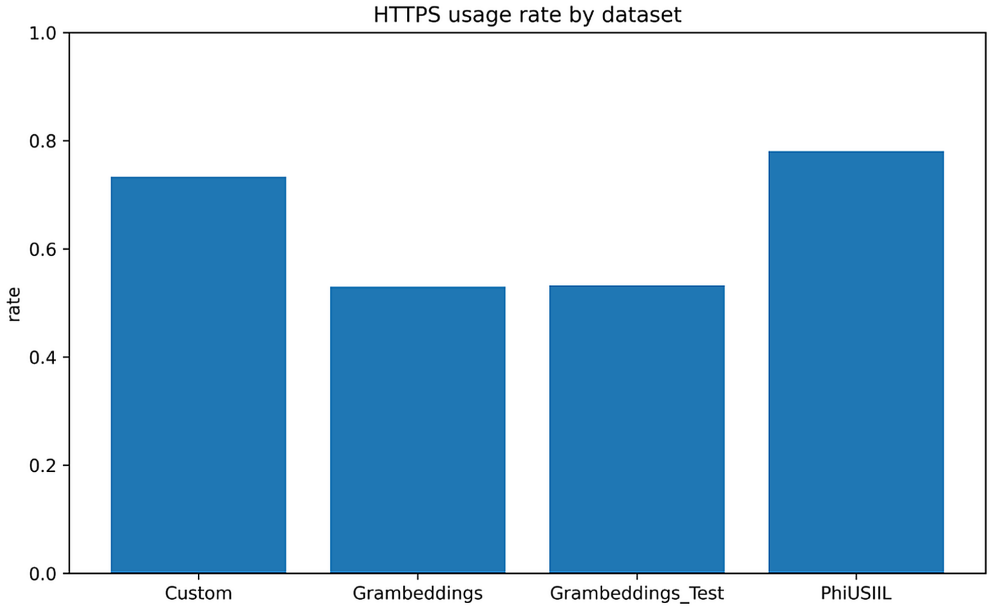



**FIGURE 2.** **HTTPS usage rate by dataset.**

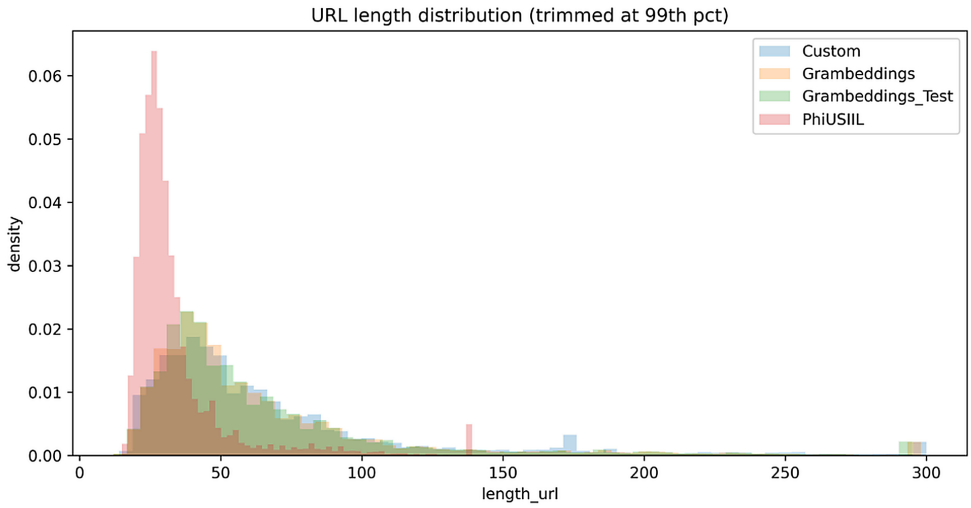



**FIGURE 3.** **URL length distribution across datasets (trimmed at 99th percentile).**

Figure 8.1: HTTPS usage and URL-length distributions across the evaluated datasets. [fig:url-length]

operational performance, so the paper reports several metrics and evaluates multiple datasets. [1]

## 8.4 Interpretation

The evidence supports three conclusions. First, the selected features can support strong detection on the custom dataset. Second, performance declines on independent datasets, meaning the detector requires continued evaluation after deployment. Third, feature importance is contextual: a signal such as HTTPS may be useful in one dataset but nearly uninformative in another. The results do not imply that all future URL populations will achieve the same scores.

# Chapter 9

# Applications

## 9.1 Potential Practical Uses

### 9.1.1 Browser-Based Protection

The X-PHIDE design targets a browser plug-in that examines external links and presents a risk category before the user proceeds. URL-focused analysis can provide an early warning without requiring the user to manually inspect technical URL characteristics. [1]

### 9.1.2 Security Awareness

The safe, suspicious, and potentially harmful categories can communicate risk in a form that is easier for users to understand than a raw model probability. Clear warnings may help users pause before entering credentials or downloading content from a suspicious destination.

### 9.1.3 Allowlist and Blocklist Support

Maintaining trusted and malicious domain lists can reduce repeated inference for known domains and allow confirmed feedback to inform subsequent decisions. Such lists need maintenance because domains can change ownership or become compromised.

### 9.1.4 Security Research and Evaluation

The paper's cross-dataset methodology can inform the evaluation of other URL classifiers. Testing on independent datasets helps identify distribution shifts and reduces the risk of reporting overly optimistic performance from one dataset alone.

## 9.2 Deployment Considerations

Real-world deployment requires attention to latency, model size, browser compatibility, false positives, false negatives, privacy, and model updates. The published demonstration is described as limited to one service and the Chrome browser. Therefore, the reported results should be treated as evidence for the method, not proof of a fully generalised commercial product.

# Chapter 10

# Advantages and Limitations

## 10.1 Advantages

- **Efficient training:** XGBoost trained in 0.95 seconds in the reported benchmark, compared with 12.01 seconds for the neural network.
- **Structured data suitability:** The method works with engineered URL and domain features and does not require a deep-learning stack or GPU for the described deployment.
- **Feature interpretation:** Feature-importance scores help examine which URL characteristics influence the model.
- **Missing-value handling:** XGBoost can handle missing values in extracted features.
- **Browser use case:** The proposed pipeline is designed for URL assessment and risk display within a browser plug-in.
- **Evaluation breadth:** Cross-dataset testing exposes limitations that a single dataset evaluation may hide.

## 10.2 Limitations

- **Dataset shift:** F1 falls from 91.16% on the custom test set to 73.25% on GramBeddings and 69.37% on PhiUSIIL.

- **Memory and inference trade-off:** XGBoost used more memory and had slower inference than the neural network in the reported comparison, despite faster training.
- **Feature dependence:** The importance of HTTPS and other signals differs across datasets.
- **Deployment scope:** The reported demonstration is limited to a single service and Chrome browser.
- **Update requirements:** The described system does not include an automatic retraining pipeline to adapt continuously to changing phishing tactics.
- **Residual risk:** No classifier can guarantee detection of every malicious URL; false positives and false negatives remain possible.

The paper's results support practical potential but also show that deployment should include monitoring, maintenance, and independent evaluation.

# Chapter 11

# Future Scope

## 11.1 Directions Discussed in the Paper

The paper identifies opportunities for improving deployment efficiency and adapting to changing URL patterns. These include model compression, automatic retraining, multi-model ensembles, and runtime validation using checks such as landing-page rendering and WHOIS/DNS lookups alongside static model predictions. [1]

## 11.2 Possible Extensions

- **Model compression:** Investigate tree pruning and export formats such as ONNX or Treelite, including quantisation where appropriate, to reduce runtime and storage requirements.
- **Automated model refresh:** Develop a monitored retraining pipeline using newly verified phishing and legitimate URLs, with safeguards against poisoned labels.
- **Ensemble detection:** Evaluate combinations of gradient-boosted trees, neural methods, and other classifiers to determine whether complementary signals improve robust performance.
- **Runtime checks:** Combine URL features with carefully selected live DNS, WHOIS, redirect, and page-level signals, while measuring the extra latency and privacy implications.

- **Broader browser support:** Test performance and usability across browsers and operating environments.
- **Continuous evaluation:** Report precision, recall, F1, false-positive rate, latency, and memory on temporally separated and independent datasets.

These are potential research and engineering extensions; they should not be interpreted as improvements already validated by the paper.

# Chapter 12

# Conclusion

Phishing URLs remain a common entry point for attacks that seek to steal information, defraud users, or distribute malware. Machine-learning-based URL screening can help identify suspicious links, but performance measured on one dataset may not transfer to other URL populations.

This paper reviewed X-PHIDE, the XGBoost-based URL phishing detection method proposed by Misiek and Hyla. The method combines allowlist and blocklist checks with engineered URL features, cross-dataset feature selection, Bayesian hyperparameter optimisation, XGBoost classification, and browser-based risk display. It uses three datasets: a custom set of 75,738 URLs, GramBeddings with 639,723 URLs, and a 47,103-URL feature-compatible subset of PhiUSIIL. [1]

The proposed system achieved 90.72% accuracy and a 91.16% F1 score on the custom test set. However, F1 scores fell to 73.25% on GramBeddings and 69.37% on PhiUSIIL, with an average of 77.93%. XGBoost trained substantially faster than the neural network in the paper's benchmark, while the neural network offered faster inference and a smaller model size. The importance of HTTPS also varied considerably between datasets.

The main lesson is that detection quality, computational efficiency, and generalisation must be considered together. Stable feature selection and independent testing are essential, but they do not eliminate all risks. A practical system should be monitored, updated as phishing patterns evolve, and assessed for false alarms and missed threats. X-PHIDE provides a useful approach to browser-based phishing detection, while its cross-dataset results clearly demonstrate the need for continued validation. [1]

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# ANNEXURE

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**CSQ413: SEMINAR**

# XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation

Date:04-09-2026


Guided By,
Prof. Mary Thresa
Associate Professor
Computer Science & Engineering

Presented By,
**Deon George**(SJC23CS086)
CSE-B
Roll No: 28





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## Course Outcome

**Course Outcomes**: After the completion of the course the student will be able to;

| CO# | CO |
|---|---|
| CO1 | Identify academic documents from the literature which are related to her/his areas of |
| CO2 | Read and apprehend an academic document from the literature which is related to her/ his areas of interest (Cognitive knowledge level: Analyse). |
| CO3 | Prepare a presentation about an academic document (Cognitive knowledge level: Create). |
| CO4 | Give a presentation about an academic document (Cognitive knowledge level: |
| CO5 | Prepare a technical report (Cognitive knowledge level: Create). |

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- Introduction
- Background
- Problem Statement
- Literature Review
- Paper Overview
- Working Principle
- Key Findings
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- Future Scope
- Conclusion
- References



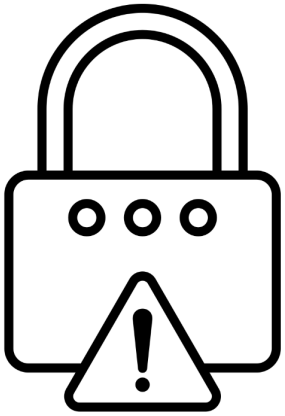



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# Introduction

- Attackers find it easier to mislead people into handing over private data, money, or approvals than to break strong technical defences — a fake URL is usually step one.
- Consequences include financial loss, reputational damage, and identity theft.
- Attack process: preparation → execution → results exploitation.
- Malicious URLs fall into several types: spam, phishing, malware, and defacement attacks.
- Most detection systems still treat this as binary: benign vs. malicious.



# Background

- Protecting users requires continuously monitoring displayed URLs and alerting on suspicious links in real time.
- Challenge 1 — Performance Optimisation: advanced ML models may need computation offloaded to a server, adding latency and cost.
- Challenge 2 — Unseen URL Detection: previously unseen phishing pages require ML/LLM-based generalisation, not static lists.
- Challenge 3 — Cross-Dataset Validation: models trained on one URL source often fail to generalise to others.
- Why XGBoost: handles imbalanced, high-dimensional tabular data; manages missing values naturally; trains faster than deep networks; and gives clear feature-importance scores.







# Problem Statement

- Models trained and tested on a single dataset can look excellent while hiding serious generalisation gaps — prior work reports F1 drops of 10–32% under dataset shift.
- RQ1: Can XGBoost achieve statistically significant cross-dataset generalisation compared to baseline methods?
- RQ2: What is the relationship between feature-importance stability and cross-dataset performance?
- RQ3: How do computational efficiency characteristics compare between XGBoost and neural networks for real-time deployment?

# Literature Review

| Sl. No. | Paper Title | Year | Main Focus | Author(s) |
|---|---|---|---|---|
| 1. | XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation (BASE PAPER) | 2026 | Presents an XGBoost-based URL phishing detection method and evaluates its generalization using cross-dataset validation. | Milosz Misiek and Tomasz Hyla |
| 2. | GramBeddings: A New Neural Network for URL Based Identification of Phishing Web Pages Through N-gram Embeddings | 2023 | Proposes a URL-only deep-learning approach using character and n-gram embeddings with CNN, LSTM and attention mechanisms. It avoids manually engineered URL features and achieves strong phishing-URL classification performance. ScienceDirect | F. Dalgıç et al. |
| 3. | A Hybrid DNN–LSTM Model for Detecting Phishing URLs | 2021 | Combines **hand-crafted lexical/NLP URL features with character-level embeddings** using DNN and LSTM/BiLSTM architectures. The hybrid approach improves phishing URL detection by exploiting both statistical URL characteristics and sequential character patterns. PubMed Central, ... +1 | A. Özcan, C. Catal, E. Dönmez and B. Şentürk |
| 4. | To Ensemble or Not: Assessing Majority Voting Strategies for Phishing Detection with Large Language Models | 2024 | Investigates LLM-based phishing detection ensembles, comparing prompt-based, model-based and hybrid majority-voting strategies. Shows that combining multiple LLM predictions can improve robustness in some situations, but the benefit depends on the relative performance of the individual models. X arXiv | F. Trad and A. Chehab |
| 5. | Phishing URL Detection Generalisation Using Unsupervised Domain Adaptation | 2024 | Investigates **cross-dataset/domain generalization** of phishing URL detectors. Shows that models can experience a **10–32% performance drop** when evaluated on a different dataset and proposes unsupervised domain adaptation to reduce this degradation. DOI +1 | F. Rashid, B. Doyle, S. C. Han and S. Seneviratne |







# Paper Overview

Paper Overview: X-PHIDE

X-PHIDE is an XGBoost-based method for detecting malicious URLs. It extends the authors' earlier conference paper by introducing cross-dataset validation, Bayesian hyperparameter optimization, and an in-depth analysis of feature importance and stability.

Datasets Used:

- Custom Dataset (75,738 URLs): Combines four sources: PhishTank, AdaURL, the Kaggle Phishing URLs Dataset by Hassaan Mustafavi, and PhishDataset.
- GramBeddings (639,723 URLs): A large-scale dataset used for feature analysis and cross-dataset evaluation.
- PhiUSIIL (47,103 URLs): An independent dataset used to evaluate the model's generalization performance.

Proposed Pipeline:

URL Input → Extraction of 90+ Features → Selection of 8 Stable Features → Bayesian Hyperparameter Optimization → XGBoost Classification → Chrome Browser Plug-in Deployment.

# Comparison

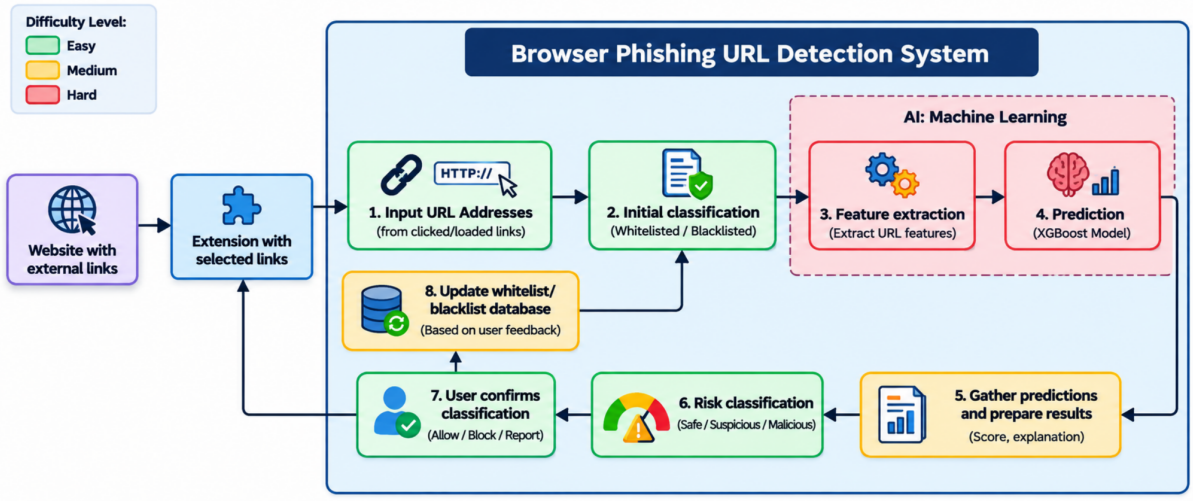
**TABLE 1. Comparison with the earlier conference version.**

| Aspect | Conference version | This extended journal version |
|---|---|---|
| Datasets | Single dataset evaluation | Cross-dataset evaluation (Custom, GramBeddings, PhiUSIIL) with held-out tests |
| Experiments | Basic train/test split on one dataset | Extensive cross-dataset validation, threshold analysis, and computational benchmarking |
| Feature selection | Within-dataset selection | Rigorous cross-dataset criteria and final set of 8 stable features |
| Hyper-parameters | Simple tuning | Bayesian optimisation and expert-guided adjustments; reported parameters included |
| Results | High single-dataset accuracy | Detailed cross-dataset metrics (average F1 77.93%), and per-dataset breakdowns |

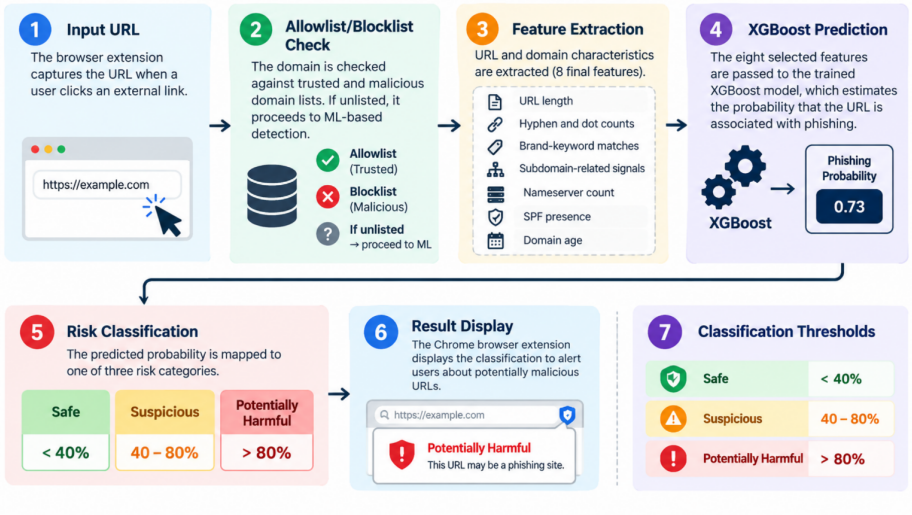




# System overview of the X-PHIDE



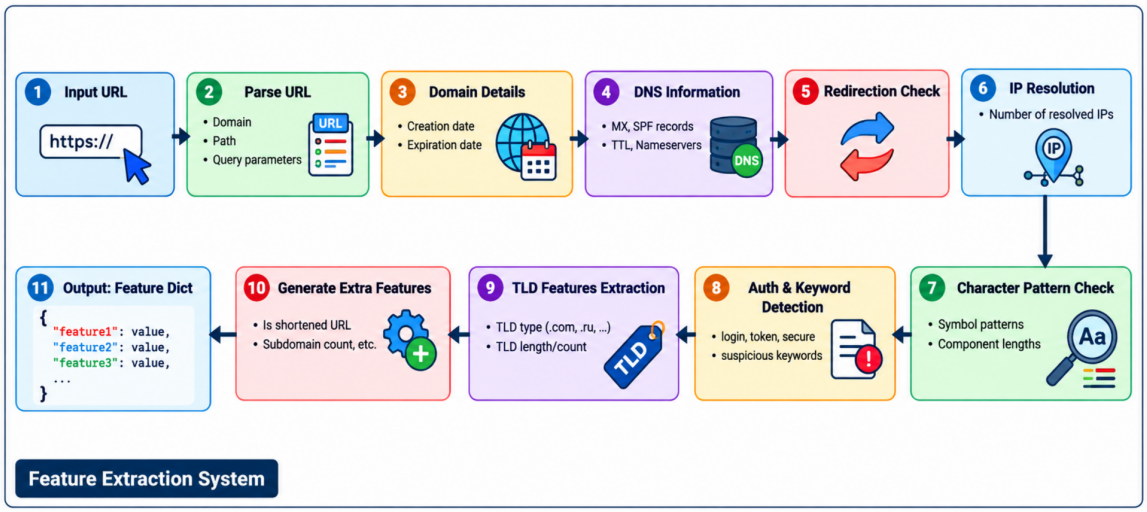
# Working Principle








# Feature extraction method



# Feature selection criterias

**TABLE 2.** Feature selection criteria.

| Criterion | Threshold | Purpose |
|---|---|---|
| Mean Importance | $> 0.02$ | Significant predictive power |
| Minimum Importance | $> 0.005$ | Always positive contribution |
| CV (Stability) | $< 0.8$ | Consistent across datasets |
| Dataset Presence | $\geq 0.75$ | Available across most datasets |

The feature selection criteria were developed through rigorous, data-driven analysis, prioritising reliability over raw performance. The four criteria work as a comprehensive evaluation framework:

- **Mean Importance $>$ 0.02:** Filters for predictive power, representing 2% of total model importance, which is significant enough to contribute meaningfully without dominating decisions.
- **Minimum Importance $>$ 0.005:** Ensures features contribute positively even in worst-case scenarios, preventing degradation of performance across datasets.
- **CV (Stability) $<$ 0.8:** Measures consistency using the coefficient of variation (CV), allowing for natural dataset variations while ensuring feature reliability.





# Final Selected 8 Features

**TABLE 7.** Final selected features.

| Feature | Type | Mean Imp. | CV | Reason |
|---|---|---|---|---|
| length_url | Lexical | 0.0319 | 0.7859 | Strong positive contributor |
| qty_hyphen_path | Lexical | 0.0315 | 0.7164 | Stable + important |
| has_brand_keyword | Security | 0.0244 | 0.5643 | Brand detection |
| subdomain_auth_count | Security | 0.0204 | 0.4807 | Auth signal |
| qty_nameservers | DNS | 0.0084 | 0.3817 | Infra signal |
| qty_dot_url | Lexical | 0.0121 | 0.3866 | URL structure |
| is_spf_domain | Security | 0.0066 | 0.5139 | SPF signal |
| time_domain_activation | Domain | 0.0125 | 0.5328 | Domain age |

**TABLE 10.** Model's cross-dataset phishing detection features.

| Feature | Description | Calculation Method |
|---|---|---|
| length_url | Total character length of the URL | Count of all characters in complete URL |
| qty_hyphen_path | Number of hyphen characters in the URL path | Count of "-" in path |
| has_brand_keyword | Binary indicator of brand names in URL | Check predefined list |
| subdomain_auth_count | Count of authentication-related keywords | Count in subdomain |
| qty_nameservers | Number of nameserver records | DNS NS record count |
| qty_dot_url | Number of dot characters in the URL | Count of "." in URL |
| is_spf_domain | SPF record presence | DNS TXT lookup |
| time_domain_activation | Domain registration date (days since epoch) | WHOIS query |

# Key Findings

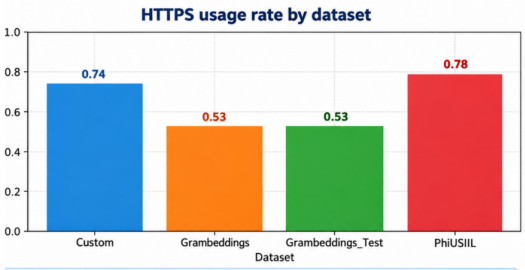
- Custom Dataset Performance: XGBoost achieved 90.72% accuracy, 94.6% recall, 86.8% specificity, 88.0% precision, and 91.16% F1 score.
- Cross-Dataset Generalisation: F1 scores were 91.16% on the Custom Test dataset, 73.25% on GramBeddings, and 69.37% on PhiUSIIL, giving an average F1 score of 77.93% (≈77.9%).
- XGBoost vs. Neural Network: XGBoost achieved 2.1% higher accuracy, trained 12.6× faster (0.95 s vs. 12.01 s), and used 3.6× less memory. The neural network offered faster inference (7.51 ms vs. 20.6 ms) and a smaller model size.
- Impact of HTTPS Feature: The performance impact varied across datasets: +4.97 percentage points (Custom), +0.04 points (GramBeddings), and +12.81 points (PhiUSIIL), indicating that the feature's contribution depends on the dataset.



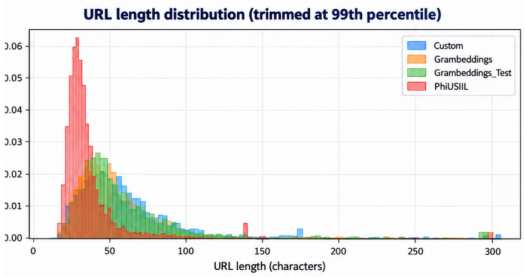
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# Key Findings



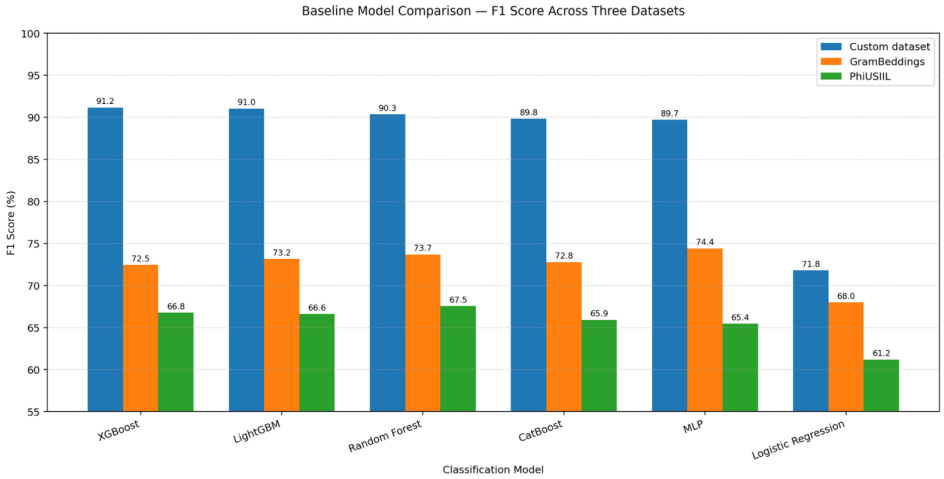

IGURE 2. HTTPS usage rate by dataset.




IGURE 3. URL length distribution across datasets (trimmed at 99th percentile).

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# Key Findings




Source: Table 12, XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation.







## Advantages

- No GPU or deep-learning stack needed for deployment
- Trains fast and handles imbalanced, tabular data well
- Built-in feature importance aids interpretability
- Robust to missing values in extracted URL features
- Real-time-suitable inference for browser plug-ins

## Limitations

- Accuracy drops under dataset shift (19.96-point gap vs. single-dataset)
- Relatively large model memory footprint for browser settings
- No caching or concurrency optimisation yet in the plug-in
- Demo currently limited to one service and the Chrome browser
- No automatic retraining pipeline against evolving adversarial URLs

# Future Scope

- Model Compression — tree pruning, ONNX/Treelite export with int8 quantisation, or distillation to shrink runtime and storage cost.
- Automatic Retraining — a pipeline to keep the model updated automatically as phishing patterns evolve.
- Multi-Model Ensembles — combining XGBoost, deep learning, and classical linear regression for the best overall detector.
- Runtime Validation — live checks such as landing-page rendering and WHOIS/DNS lookups alongside the static model.



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# Conclusion

- XGBoost offers efficient training, strong predictive performance on structured URL features, and feature-importance scores that make its decisions explainable.
- These properties make it well suited for real-time, browser-based security applications where low latency matters.
- Cross-dataset experiments expose real limits: performance shifts with data-collection differences, and attackers keep adapting obfuscation techniques.
- Real-world deployment needs continuous monitoring, regular retraining, and additional runtime safeguards.
- **Takeaway: speed and interpretability make XGBoost the practical choice — but not yet a finished product.**

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